

Sentiment Analysis on Amazon Fine Food Reviews

Internship Project – Week 1

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INTRODUCTION

This project focuses on analyzing customer reviews from the Amazon Fine Food Reviews dataset using Natural Language Processing (NLP). The objective is to classify customer reviews into Positive, Negative, and Neutral sentiments using the TextBlob library. Sentiment analysis helps businesses understand customer opinions, measure customer satisfaction, and identify areas for improvement. The project also includes data visualization to present the findings in a clear and meaningful way.

DATASET INFORMATION

The dataset used for this project is the **Amazon Fine Food Reviews** dataset obtained from Kaggle. The first **5,000 reviews** were selected for analysis. After data cleaning, which involved removing missing and duplicate reviews, **4,984 valid reviews** remained. The analysis primarily used two columns: **Text** (customer review) and **Score** (customer rating).

METHODOLOGY

The dataset was loaded using the Pandas library and explored to understand its structure. Data cleaning was performed by removing missing and duplicate reviews while retaining only the required columns. The TextBlob library was then used to calculate the sentiment polarity of each review. Based on the polarity score, reviews were classified as Positive, Negative, or Neutral. Finally, various visualizations were created using Matplotlib and Seaborn to analyze the distribution of sentiments and customer ratings.

RESULTS

The sentiment analysis revealed that **88.34%** of the reviews were classified as **Positive**, **10.05%** as **Negative**, and **1.61%** as **Neutral**. The majority of customers expressed positive opinions about the products, indicating a high level of customer satisfaction. The rating distribution also showed that most customers gave 4-star and 5-star ratings. Negative reviews mainly reflected dissatisfaction with product quality, packaging, taste, or delivery experience.

RECOMMENDATION

Based on the analysis, the business should continuously monitor customer feedback, especially negative reviews, to identify recurring issues and improve product quality and customer experience. Maintaining product consistency, improving packaging, and addressing customer concerns promptly can further increase customer satisfaction and encourage more positive reviews.

CONCLUSION

This project demonstrates how Natural Language Processing techniques can be applied to analyze customer reviews effectively. By combining data cleaning, sentiment analysis, and visualization, meaningful insights were extracted from customer feedback. The results show that most customers were satisfied with the products, while the identified negative feedback provides valuable information that businesses can use to improve their products and services. Overall, sentiment analysis is an effective tool for understanding customer opinions and supporting data-driven business decisions.